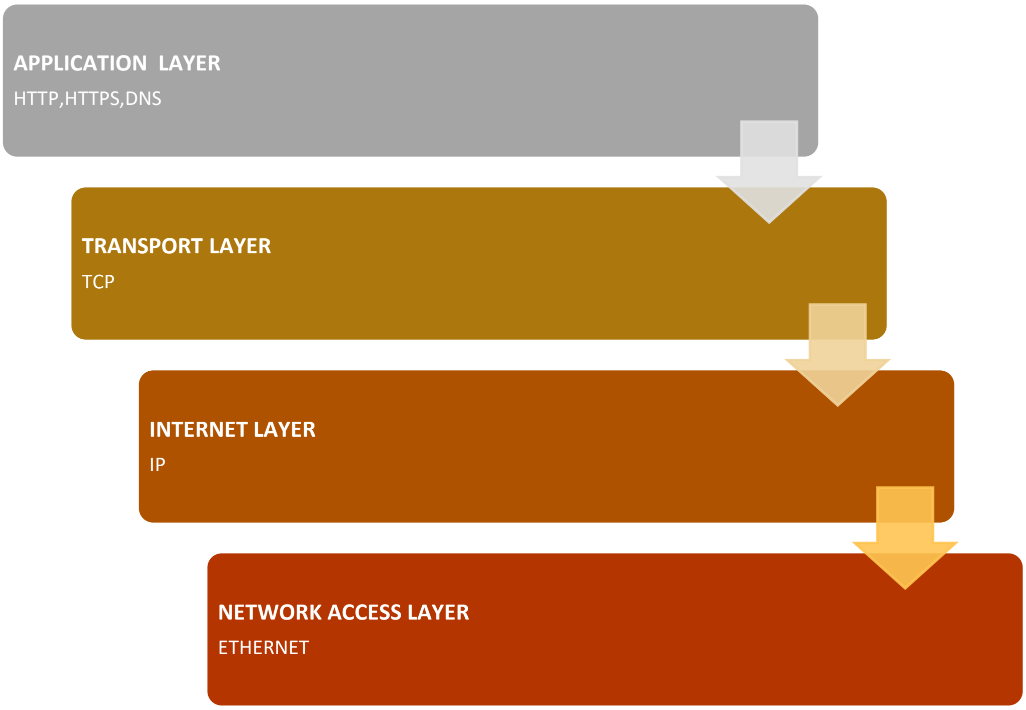
How the Internet Delivers a Web Page

Website Chosen: Amazon India (www.amazon.in)

This report explains how the Internet loads the Amazon India homepage when a user enters www.amazon.in into a web browser. It describes the role of networking protocols such as **DNS, TCP, and HTTP/HTTPS** in finding the server, establishing a connection, and delivering the webpage to the user. The report also explains how these protocols work together to provide a fast and reliable browsing experience.

1.TCP/IP Protocol Stack and the Role of Each Layer

The TCP/IP protocol stack consists of four layers. Each layer performs a specific function in delivering data from Amazon's web server to the user's browser.



a) Application Layer

The Application Layer provides network services directly to user applications. In the case of Amazon, the web browser uses HTTPS to communicate securely with Amazon's web server. DNS also operates at this layer to convert the domain name into an IP address.

b) Transport Layer

The Transport Layer uses TCP to establish a reliable connection between the browser and Amazon's server. TCP divides data into smaller segments and ensures that every segment reaches its destination correctly.

c) Internet Layer

The Internet Layer uses the Internet Protocol (IP) to route packets from the user's device to Amazon's servers through multiple routers across the Internet.

d) Network Access Layer

The Network Access Layer is responsible for transmitting data over the physical network using technologies such as Wi-Fi or Ethernet.

2. Role of DNS in Converting the Domain Name into an IP Address

Computers communicate using IP addresses rather than domain names. When a user enters **www.amazon.in**, the browser must first determine the corresponding IP address.

The Domain Name System (DNS) performs this translation.

Steps in DNS Resolution

1. The user types **www.amazon.in** into the browser.
2. The browser checks its local DNS cache.
3. If the address is not found, the request is sent to the operating system's DNS cache.
4. If still unavailable, the request is forwarded to the ISP's DNS resolver.
5. The resolver contacts the Root DNS Server.
6. The Root Server directs the query to the **.in** Top-Level Domain (TLD) server.
7. The TLD server identifies Amazon's authoritative DNS server.
8. The authoritative server returns the IP address of **www.amazon.in**.
9. The browser stores the IP address in its cache for future use.

4. How HTTP is Used to Request and Receive the Webpage

HTTP (HyperText Transfer Protocol) is the protocol used to transfer web pages between a browser and a web server. Amazon uses the secure version, HTTPS, which encrypts communication using TLS to protect sensitive information such as passwords and payment details. When a user opens **www.amazon.in**, the browser sends an HTTPS GET request to Amazon's web server. The server processes the request and returns an HTTP response containing the HTML document, images, CSS files, JavaScript files, and other resources needed to display the webpage.

Process

1. Browser sends HTTPS request.

2. Amazon server receives the request.
3. Server processes the request.
4. Server returns HTML, CSS, JavaScript, images, and product information.
5. Browser renders the complete Amazon homepage.

4. How TCP Provides Reliable Communication for HTTP

TCP provides reliable communication by ensuring that data is delivered correctly and in the correct order. Before sending data, TCP establishes a connection between the browser and the Amazon server. It divides the webpage into smaller packets and assigns sequence numbers to them. The receiver sends acknowledgements (ACKs) after receiving the packets. If any packet is lost or damaged, TCP retransmits it. TCP also performs error checking and flow control to prevent data loss. This ensures that the Amazon webpage loads completely and accurately.

5. Client–Server Architecture of Amazon

Amazon follows the **Client–Server architecture**. In this architecture, the user's web browser acts as the **client**, and Amazon's computers act as the **server**. When the user enters **www.amazon.in**, the browser sends a request to the Amazon server. The server processes the request and sends the requested webpage back to the browser.

Amazon uses the Client–Server architecture because all product information, customer accounts, and order details are stored on centralized servers that provide services to millions of users.

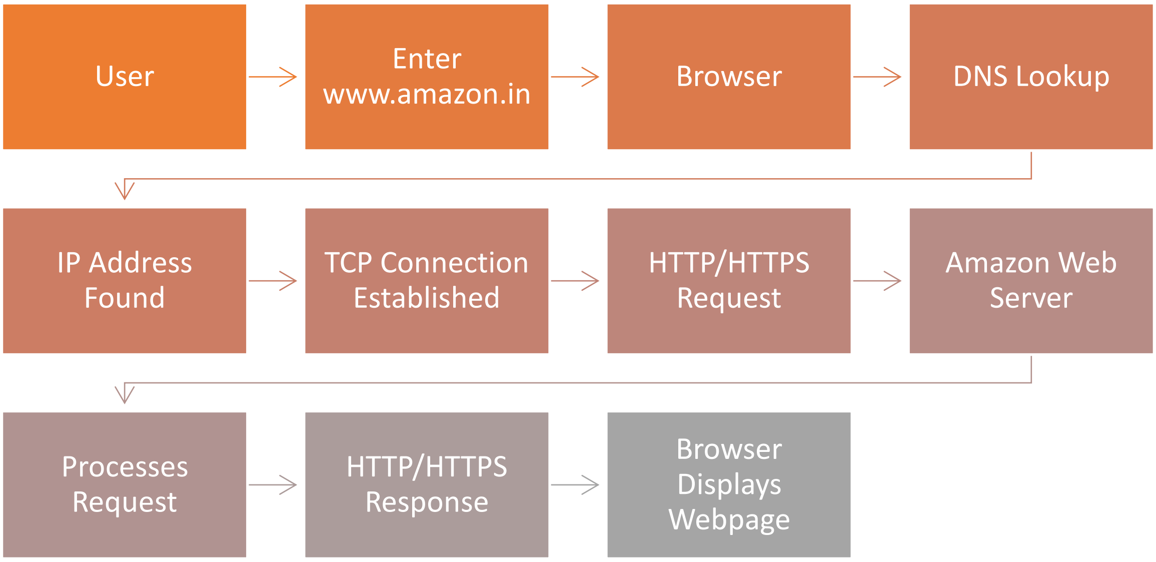
6. Comparison between HTTP and FTP

HTTP	**FTP**
Used to access web pages	Used to transfer files
Uses Port 80 (HTTP) or 443 (HTTPS)	Uses Port 21
Used by web browsers	Used by FTP clients
Transfers web content such as HTML, CSS, and images	Transfers files such as documents, videos, and software

7. Role of Cookies in Improving User Experience

Cookies are small text files stored by the browser that remember information about the user. They improve the user experience by saving login details, language preferences, and shopping cart items.
When a user logs into Amazon and adds products to the shopping cart, cookies store this information. If the user visits Amazon again later, the website remembers the login session and the items in the shopping cart, making shopping faster and more convenient.

8. Flow Diagram of Communication Between the Browser and the Web Server



9.Conclusion

Application-layer protocols play an important role in our everyday use of the Internet. They allow users to access websites, send emails, transfer files, and use many online services quickly and efficiently. In the case of **Amazon India (www.amazon.in)**, the **DNS** protocol converts the website name into an IP address so that the browser can locate the correct server. **TCP** establishes a reliable connection and ensures that all data is delivered accurately without loss. **HTTP/HTTPS** is used to request and receive the webpage, while **HTTPS** also provides encryption to protect sensitive information such as login credentials and payment details. Cookies improve the user experience by remembering user preferences, login sessions, and shopping cart items. Together, these protocols work seamlessly to provide a fast, reliable, and secure browsing experience. Understanding the role of application-layer protocols helps us appreciate how modern websites and online services function efficiently in our daily lives.